

Data Warehousing and Data Mining

CSIT — 7th Semester — 2078 — Answer Sheet
Subject Code: CSC410

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

1. Write down any one advantage and disadvantage of MOLAP over ROLAP. Define signed network and how do you check whether it is balanced or not? How beam search reduces the space complexity? Illustrate with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. How concept hierarchy is used in extracting information? Generate the frequent pattern from the following data set using FP growth, where minimum support=3. T_ID Items 1A,B,C,D,F,H 2A,D,E,F 3C,D,F 4B,H 5A,C,F,G,H 6C,D,E,G 7A,C,D,I

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. How do you compare two classifiers? Given the points A(3,7), B(4,6), C(5,5), D(6,4), E(7,3), F(6,2), G(7,2) and (8,4), find the core points, border points and outliers using DBSCAN. Take Eps = 2.5 and MinPts = 3.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Attempt any Eight questions [8×5=40]

4. When a pattern is said to be interesting? List the issues of data mining.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Define data discretization. Describe the tasks for data preprocessing.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. Define spatial data mining. What are the challenges of multimedia mining? Describe with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Consider the following data Set.

Confident	Studied	Sick	Result	Yes	No	No	Fail	Yes	No	Yes	Pass	No	Yes	Yes	Fail
No	Yes	No	Pass	Yes	Yes	Yes	Pass	Find out whether the object with attribute Confident = Yes, Studied = Yes, Sick = No will Fail or Pass using Bayesian classification.							

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. What are the choices for data cube materialization? Explain the strategies for cube computation.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Show the conflict between theory of balance and status. How do you improve Apriori?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Differentiate between star schema and snow flake schema. List any two methods for data normalization.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. How do you evaluate the accuracy of a classifier? Discuss the advantages of using K-fold cross validation.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Apply K(=2)- Means algorithm over the data (185,72), (170, 56), (168,60), (179,68), (182,72), (188,77) up to two iterations and show the clusters. Initially choose the first two objects as initial centroids.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.